Compiler Design

Lecture-03

Introduction to Grammars

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Grammars

- Grammar is a formal system that defines a set of rules for generating valid strings within a language.
- Grammar is Composed of **two** basic elements:
    - Terminal Symbols :Terminal symbols are those that are the components of the sentences generated using grammar and are represented using small case letters like a, b, c, etc.

    - Non-terminal Symbols : Non-terminal symbols are those symbols that take part in the generation of the sentence but are not the component of the sentence.These symbols are represented using a capital letters like A, B, C, etc.

Representation of Grammar

- Any Grammar Can be represented with 4 tuples.
  <N,T,P,S>

- N - Finite Non-Empty Set of Non-Terminal Symbols.
- T - Finite Set of Terminal Symbols.
- P - Finite Non-Empty Set of Production Rules.
- S - Start Symbol (Symbol from where we start producing our sentences or strings).

Production Rules

- A production or production rule in computer science is a rewrite rule specifying a symbol substitution that can be recursively performed to generate new symbol sequences. It is of the form α-> β where α is a Non-Terminal Symbol which can be replaced by β which is a string of Terminal Symbols or Non-Terminal Symbols.

Example

Grammar G1 = <N, T, P, S>

T = {a,b}   #Set of terminal symbols

1 2 3 4 5

P = { A -> Aa, A -> Ab, A -> a ,A -> b, A -> ε}   #Set of all production rules

S = {A}   #Start Symbol

<u>Derivation of Strings :</u>

A->a   #using production rule 3

OR

A->Aa   #using production rule 1

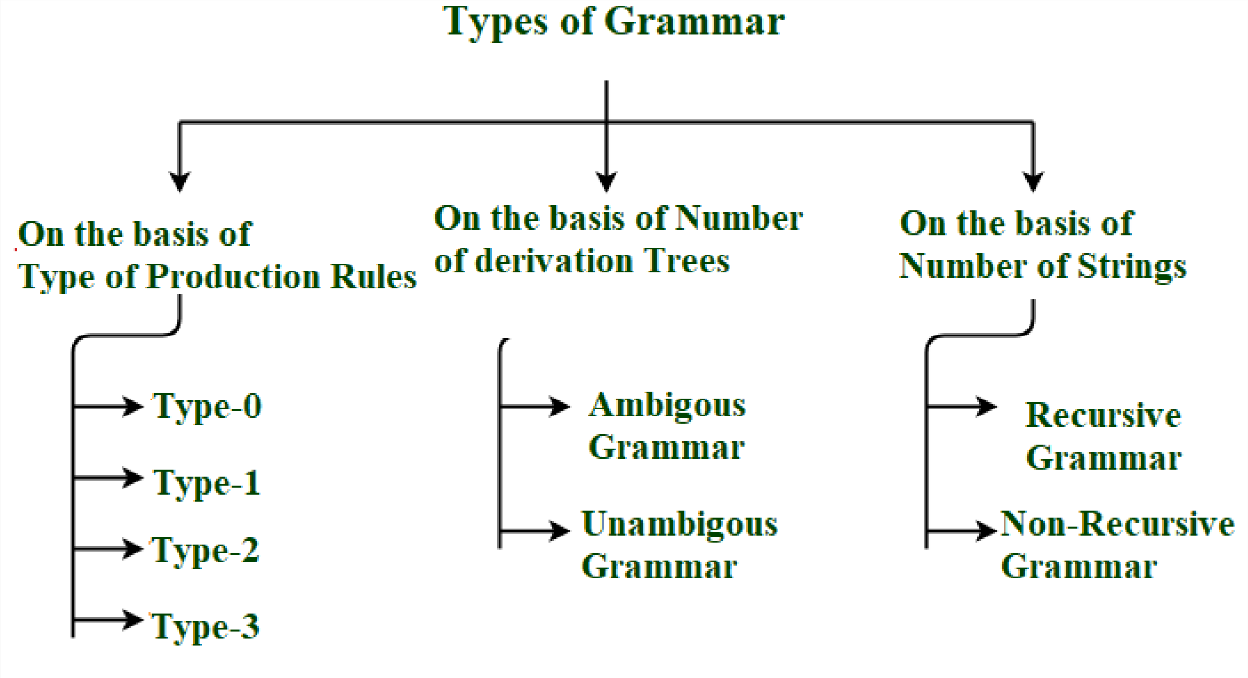
Aa->ba   #using production rule 4

OR

A->Aa   #using production rule 1

Aa->AAa   #using production rule 1

Classification of Grammar



Type-0

Unrestricted Grammar

- No restriction on production rules
- Rule format:

$$\alpha \rightarrow \beta$$

where α and β are any strings of terminals and non-terminals
(α must contain at least one non-terminal)

Example:

$AB \rightarrow BA$
$S \rightarrow aSb$

Type-1

Context Sensitive Grammar (CSG)

Rule Format

$\alpha \; A \; \beta \rightarrow \alpha \; \gamma \; \beta$

- $A \longrightarrow$ single non-terminal
- $\gamma \neq \varepsilon$
- Length of RHS ≥ LHS

Example:

G:

P: $Ac \rightarrow Bbcc$

$\alpha A \beta \rightarrow \alpha \gamma \beta$

$\varepsilon Ac \rightarrow \varepsilon Bbcc$

Type-2

Context Free Grammars

Rule Format

$A \rightarrow \gamma$

- A = single non-terminal
- γ = terminals and/or non-terminals

Example: $S \rightarrow aSb \mid \varepsilon$

Generates : ε

ab

aabb

aaabbb

Type-3

Regular Grammar

Right Linear :

$A \rightarrow aB$

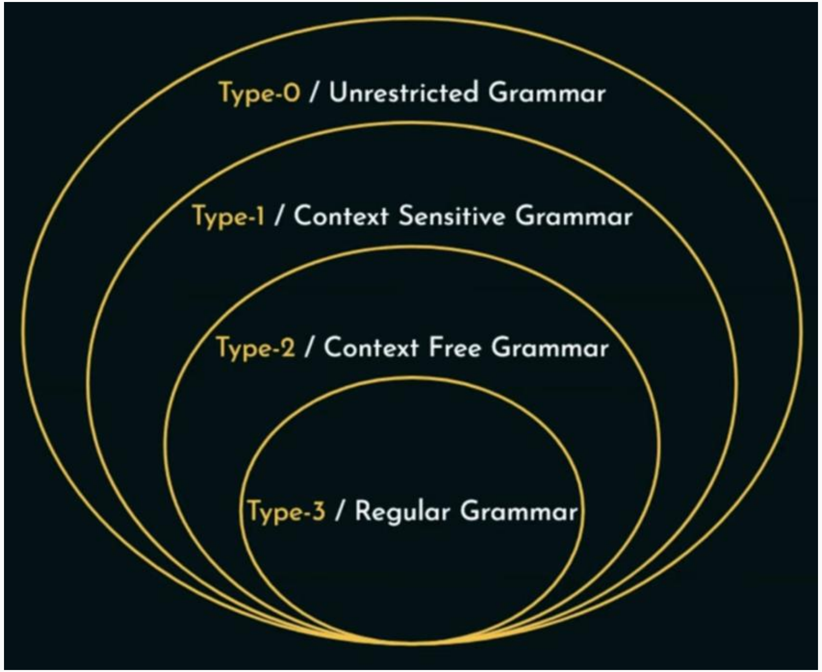
$A \rightarrow a$

Left Linear :

$A \rightarrow Ba$

$A \rightarrow a$

Chomsky Hierarchy



Thank You